Evaluation of Carbon Fiber and Aerogel for Increased Dexterity in Firefighter Turnout Gloves

Introduction

Firefighters operate in high temperature, low visibility, and oxygen limited environments where hand performance is critical to **survival** and **task success**. While modern turnout gloves provide strong thermal and cut protection, they are often **thick**, **stiff**, and **restrictive**, reducing manual dexterity by over 50% compared to bare hands during task performance (McLellan & Selkirk 2004).

Figure 1: Current Generation Firefighting Gloves (Fire Safety USA)

This loss of fine motor ability forces firefighters to use more effort for basic tasks, increasing **fatigue** and **oxygen consumption** during emergency operations. Improving hand mobility without compromising protection is therefore a major **engineering challenge** in modern firefighting gear.

Purpose

This new firefighter glove composition integrates a **carbon fiber weave** and **Aerogel fabric** into a **Nomex-base layer**, resulting in a lighter, stronger, and more flexible structure. These materials also allow for **greater flexibility** and **improved tactile sensitivity**, crucial for performing fine motor tasks. High strength composite materials are used to provide enhanced cut resistance, abrasion protection, and structural support. By **reducing bulk**, **stiffness**, and **internal friction**, this design aims to decrease hand fatigue, lower oxygen demand, and increase task efficiency. The prototype glove's **improved mobility** and **comfort** enhance a firefighter's ability to perform in extreme environments, ultimately making them **faster**, **safer**, and **more effective**.

Methods

Phase One: Construction Sample Creation

Swatches of **Carbon Fiber Weave**, **Aerogel Fabric,** and **Nomex** were cut to size and bonded together via **Flame Retardant HTV**.

Figure 2: Prepped Pieces of Protoype Construction (Photo taken by Finalist, 2026)

Figure 3: Bonding Layers together via Heat Press (Photo taken by Finalist, 2026)

The **completed construction** was then folded over on itself and stapled at the edges to create a **pocket** in order to simulate a glove.

Figure 4: Bonded Construction ready for folding (Photo taken by Finalist, 2026)

Figure 5: Completed Prototype "Pocket" for Testing (Photo taken by Finalist, 2026)

Phase Two: Construction Sample FPP Testing

A **Brick Oven** was lit with firewood in order to simulate a **flashed over structure**, holding average temperatures at **1500 F°** for test.

Figure 6: Flashover simulation started in brick oven (Photo taken by Finalist, 2026)

Figure 7: Measuring temperature of simulation for test (Photo taken by Finalist, 2026)

The **prototype sample** and **a standard fire glove** were placed inside the oven for **15 seconds**, with data points collected every **3**.

Figure 8: FPP Testing prototype construction (Photo taken by Finalist, 2026)

Figure 9: FPP Testing control glove (Photo taken by Finalist, 2026)

Results

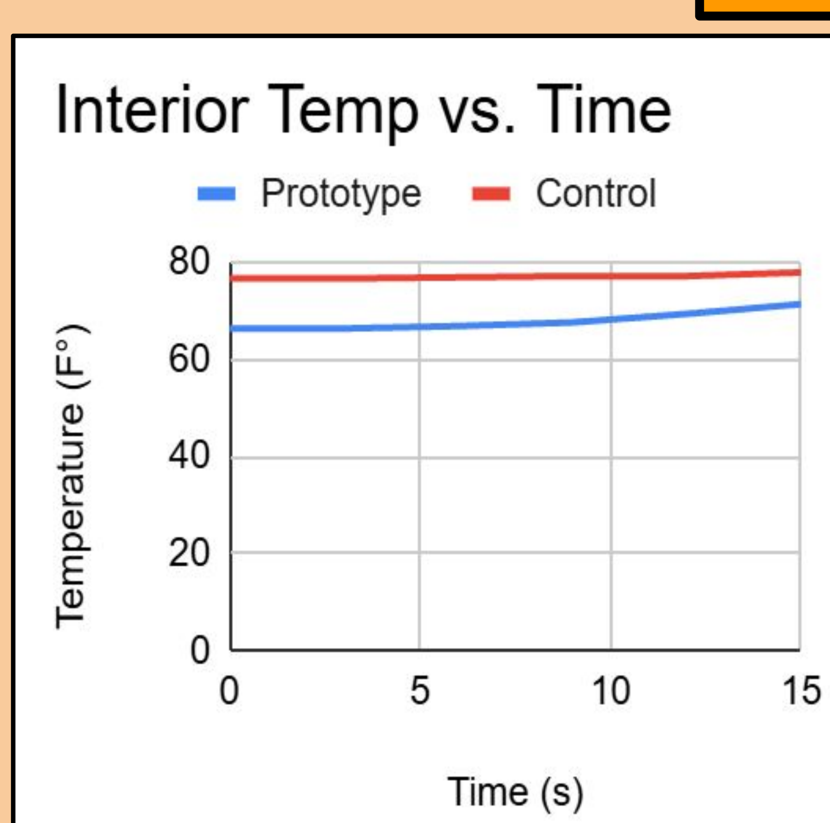

Figure 10: Graph displaying interior temperature of both prototype construction and control glove over time.(Graph created by Finalist using Google Sheets, 2026)

After completing four trials, the interior temperature of the prototype increased by **8 °F** on average per trial, blocking out the **majority** of the heat. It was also observed that the prototype construction **dissipated exterior heat faster** than the control, becoming **cool to the touch after just seconds** compared to minutes respectively.

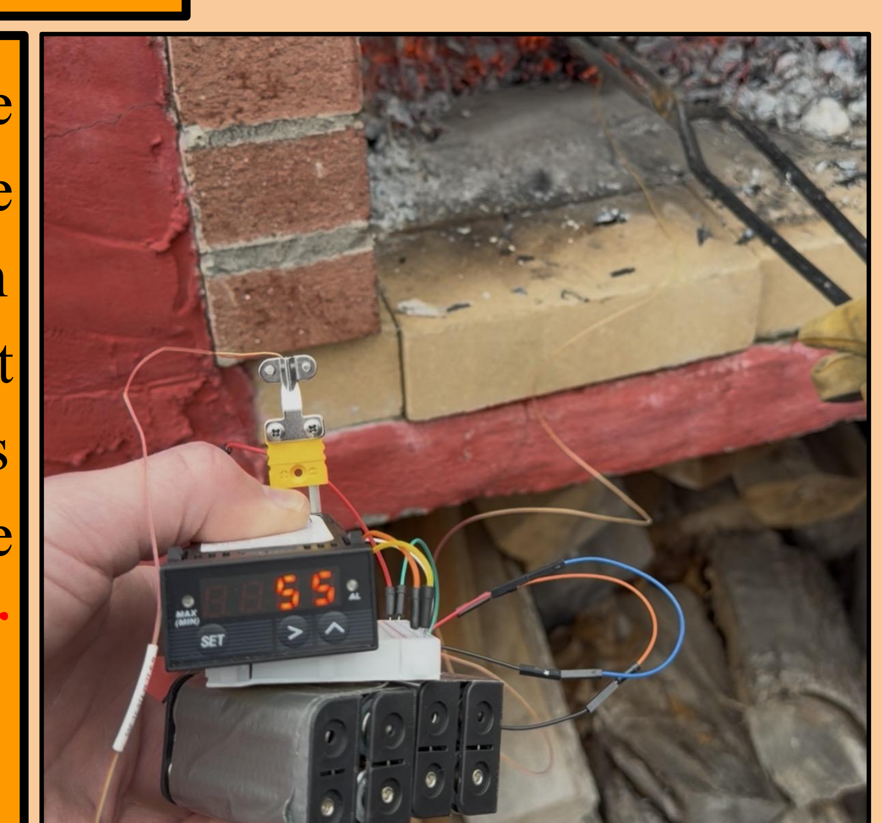

Figure 11: K-Type Thermocouple Setup used to collect interior temperature data (Photo taken by Finalist, 2026)

Conclusions

The prototype construction demonstrated **consistent** performance under high temperature exposure, remaining **unharmed** after all trials. In contrast, the control glove **failed after only four trials,** burning and becoming stiff.

Figure 12: Control Glove failure after fourth trial (Photo taken by Finalist, 2026)

These results suggest that the prototype construction provides **superior thermal protection**, **increased structural integrity**, and **greater durability**, compared to the current generation of firefighting gloves. The **thinness** and **flexibility** make it a viable alternative for a **more dextile** fire glove.

Applications

This new material construction, featuring **rapid heat dissipation** and **high flexibility**, has the potential to significantly improve **firefighter turnout gear**, **industrial protective clothing**, and **motorsport suits**. Its combination of heat resistance, durability, and enhanced mobility offers **substantial advantages** in high heat environments, ensuring **protection**, **comfort**, and improved performance for users in **extreme conditions**, while maintaining long-term wearability, reducing fatigue, and allowing for **greater maneuverability** during **critical tasks** and **prolonged exposure**.

References

1) Hexcel Corporation. (n.d.). Carbon fiber reinforcement properties and applications. https://www.hexcel.com/
2) McLellan, T. M., & Selkirk, G. A. (2004). The management of heat stress for the firefighter: A review of work conducted on behalf of the Toronto Fire Service. Industrial Health, 42(4), 414–426.
3) National Fire Protection Association. (2023). NFPA 1971: Standard on protective ensembles for structural fire fighting and proximity fire fighting. NFPA.